

Study Questions: The Testing Agent

Analytical Questions

1. Why is the transition from creator to evaluator cognitively difficult for inventors?
Explain using the concepts of prior beliefs, precision weighting, and prediction error.
2. What is the inventor's "observation model" and how does it differ from the observation model of a naive user? Give specific examples of what each agent attends to and ignores.
3. Explain how confirmation bias arises as a natural consequence of Bayesian inference with strong priors. Why does this make self-testing unreliable?
4. What is the difference between high sensory precision and low sensory precision in the context of prototype testing? How does each affect the inventor's ability to update their beliefs?
5. Why are beta testers valuable as "distributed sensors"? What properties of a beta tester population maximize the diversity of prediction errors?
6. Define adversarial testing in Active Inference terms. How does an adversarial tester's generative model differ from a standard user tester's?
7. Explain how sunk cost bias functions as a precision problem in Active Inference. What is the agent weighting too heavily, and what should it weight more?
8. What is pre-registration in testing, and why does it reduce the space for post-hoc rationalization? How does it constrain the agent's degrees of freedom?
9. How does expertise create blind spots in testing? Use the concept of "precision-weighted attention" to explain why experts sometimes cannot see what novices see.
10. Compare the observation models of three different testing agents: the inventor, a domain expert, and a child. What unique prediction errors does each generate?

Applied Questions

1. You have invented a new type of jar opener and are convinced it works perfectly. Design a testing protocol that includes at least three types of testing agents, specifying what each agent is likely to see that you cannot.
2. A startup founder has tested their app exclusively with friends and family, who all report that it is "great." Using Active Inference concepts, explain why this testing strategy is flawed and propose improvements.
3. You are watching a user test your prototype and they are using it completely wrong. You feel a strong urge to intervene and show them the "right way." Analyze this urge in terms of prediction error and prior beliefs. Should you intervene? Why or why not?
4. An automotive company is testing a new dashboard interface. Design an adversarial testing protocol that would reveal failure modes not found in standard usability testing.
5. Two testers evaluate your prototype. Tester A (an engineer) says it works well but the materials are wrong. Tester B (a potential customer) says the materials are fine but they do not understand what the product does. How do you reconcile these conflicting observations using the concept of different generative models?
6. You are developing a medical device and must decide between testing with 5 expert physicians or 50 random adults. Analyze this trade-off using the concepts of agent expertise, observational precision, and coverage of the prediction error space.
7. A children's toy company discovers that their toys, designed by adults, consistently fail to engage children under 5. Using the testing agent framework, explain this pattern and propose a testing strategy that addresses it.
8. You notice that your first test always seems to go either very well or very poorly, and this initial result colors your interpretation of all subsequent tests. Identify the bias at work, explain its mechanism, and propose a countermeasure.

9. Design a "blind testing" protocol for a physical product (not software). What information do you withhold from the tester? What information do you provide? How do you prevent the tester's expectations from being shaped by the testing setup itself?
10. Your co-inventor disagrees with your interpretation of test results. They believe the prototype performed well; you believe it failed. Using the concept of different observation models, analyze how two people can observe the same test and reach opposite conclusions. How do you resolve this?